

A User-Centric Business Intelligence Framework for Predictive Customer Churn in E-Commerce

Researcher: Muhammad Tehmas (25MEIT003)

Institution: Mehran University of Engineering and Technology (MUET)

Supervisors: Prof. Dr. Shah Nawaz Talpur | Co-Supervisor: Engr. Madeha Memon

Current Phase: Phase 1 Completed (Data Preprocessing & EDA)

1. Phase 1 Executive Summary & Dataset Profile

In this foundational phase, the baseline academic dataset (Kaggle E-Commerce Customer Churn) was ingested, cleaned, and prepared for interpretable machine learning modeling. A thorough exploratory data analysis (EDA) was conducted to identify data quality issues, missing distributions, and class imbalances.

Metric / Parameter	Value / Description
Dataset Source	Kaggle E-Commerce Customer Churn Dataset (Phase 1 Baseline)
Total Customer Records	5,630 customers
Predictive Features	18 behavioral & categorical features
Target Variable	Churn (1 = Churned, 0 = Retained)
Overall Churn Rate	16.84% (Imbalanced Class Distribution)

2. Methodological Decision Rationale (Why We Did What We Did)

Decision 1: Why Median Imputation for Missing Values?

During inspection, 7 continuous variables showed missing values (e.g., DaySinceLastOrder had 307 missing, Tenure had 264 missing). In e-commerce customer behavior data, metrics are heavily right-skewed due to outliers (power buyers or extreme distances). If we used Mean (average) imputation, extreme outliers would pull the average upwards, distorting reality. **Median imputation (50th percentile)** was selected because it is robust against outliers and perfectly represents the typical customer's behavior.

Decision 2: Why Did We NOT Use One-Hot Encoding?

Traditional machine learning pipelines convert categorical strings (like 'PreferredLoginDevice = Mobile') into multiple binary columns (e.g., 'Device_Mobile_1', 'Device_Laptop_0'). However, our primary thesis objective is **User-Centric Business Intelligence** for non-technical SME managers. We chose to preserve categorical strings as raw text because our Phase 2 model—**Explainable Boosting Machines (EBM)**—handles text natively and produces 100% human-readable rules. One-Hot Encoding would destroy interpretability.

Decision 3: Why 80/20 Stratified Train-Test Splitting?

Our dataset exhibits a natural class imbalance (~16.84% Churn vs 83.16% Retained). A simple random train-test split could accidentally assign an uneven percentage of churners to the training set. By using **Stratification (stratify=y)**, we guarantee that exact ~16.84% churn ratio is mirrored across both Train and Test splits, preventing sampling bias and ensuring fair model evaluation.

3. Preprocessing & Imputation Log

Feature Name	Missing Count	Imputed Median Value
Tenure	264	9.0
WarehouseToHome	251	14.0
HourSpendOnApp	255	3.0
OrderAmountHikeFromlastYear	265	15.0
CouponUsed	256	1.0
OrderCount	258	2.0
DaySinceLastOrder	307	3.0

4. Remaining Thesis Roadmap (Phases 2 to 5)

- **Phase 2 (Core Modeling & Comparative Analysis):** Train Microsoft's Explainable Boosting Machine (EBM) on X_train. Compare accuracy and ROC-AUC against a baseline Random Forest model to validate performance.
- **Phase 3 (Explainability Export & BI Dashboard):** Export global feature contributions and customer risk probabilities to structured CSVs. Design custom minimal backgrounds in Figma and build the interactive Power BI dashboard.
- **Phase 4 (Real-World Local Validation):** Ingest raw logistics and sales records from Urban Daraz. Perform RFM (Recency, Frequency, Monetary) feature engineering and stress-test the framework.
- **Phase 5 (Usability Evaluation & Defense):** Conduct System Usability Scale (SUS) and Technology Acceptance Model (TAM) surveys with SME managers. Finalize MUET thesis chapters for submission.